

Busca Bibliográfica na Saúde: Integrando Bases de Dados Clássicas e Ferramentas de IA

**Programa de Pós-graduação Interdisciplinar em Ciências da
Saúde – Campus Baixada Santista**

Prof. Dr. Rafael Herling Lambertucci

Quando precisamos buscar conhecimento?

Redação científica



Desenvolvimento profissional





Categoria (Objetivo)	Tipo de material	Descrição resumida	Nível acadêmico mais comum
Formação	<i>Projeto de pesquisa</i>	<i>Documento inicial que apresenta o plano da investigação.</i>	<i>Graduação, pós-graduação e pesquisadores</i>
	<i>TCC (Trabalho de Conclusão de Curso)</i>	<i>Demonstra autonomia e aplicação de conhecimentos.</i>	<i>Graduação</i>
	<i>Monografia</i>	<i>Estudo aprofundado sobre tema específico.</i>	<i>Graduação, especialização</i>
	<i>Dissertação</i>	<i>Contribuição aplicada ou metodológica.</i>	<i>Mestrado</i>
	<i>Tese</i>	<i>Contribuição original e significativa para a ciência.</i>	<i>Doutorado</i>
Avaliação	<i>Ensaio acadêmico</i>	<i>Texto reflexivo e argumentativo.</i>	<i>Graduação, pós-graduação</i>
	<i>Resenha crítica</i>	<i>Avaliação de obra científica ou cultural.</i>	<i>Graduação, pós-graduação</i>
	<i>Fichamento</i>	<i>Registro sistemático de leituras.</i>	<i>Graduação, pós-graduação</i>
	<i>Revisão aos pares</i>	<i>Avaliação de trabalhos científicos (projetos e artigos)</i>	<i>Pós-graduação, pesquisadores</i>
Divulgação científica	<i>Artigo científico</i>	<i>Publicação formal em periódicos.</i>	<i>Graduação (iniciação científica), pós e pesquisadores</i>
	<i>Resumo expandido</i>	<i>Apresentação sintética em congressos.</i>	<i>Graduação, pós-graduação e pesquisadores</i>
	<i>Pôster científico</i>	<i>Exposição visual de resultados em eventos.</i>	<i>Graduação, pós-graduação e pesquisadores</i>
	<i>Relatório técnico/científico</i>	<i>Documento aplicado com descrição detalhada de métodos e resultados.</i>	<i>Graduação e pós-graduação (bolsistas), e pesquisadores (relatórios de projetos)</i>

Major type	Examples
Primary or original research articles	Randomized controlled trial Clinical trial Before-and-after study Cohort study Case-control study Cross-sectional survey Diagnostic test assessment Case report/case series Technical note
Secondary or review articles	Narrative review article Systematic review Meta-analysis
Special articles	Letters/correspondence Short communications Editorials/opinion Commentaries Pictorial essay Other special categories
Tertiary literature	Textbooks, handbooks, manuals Trade or professional publication articles Encyclopedias
Gray literature	Conference proceedings, posters, abstracts Government reports For-profit and nonprofit organization reports online forums Blogs, microblogs, tweetchats, and other social media

Various Types of Scientific Articles

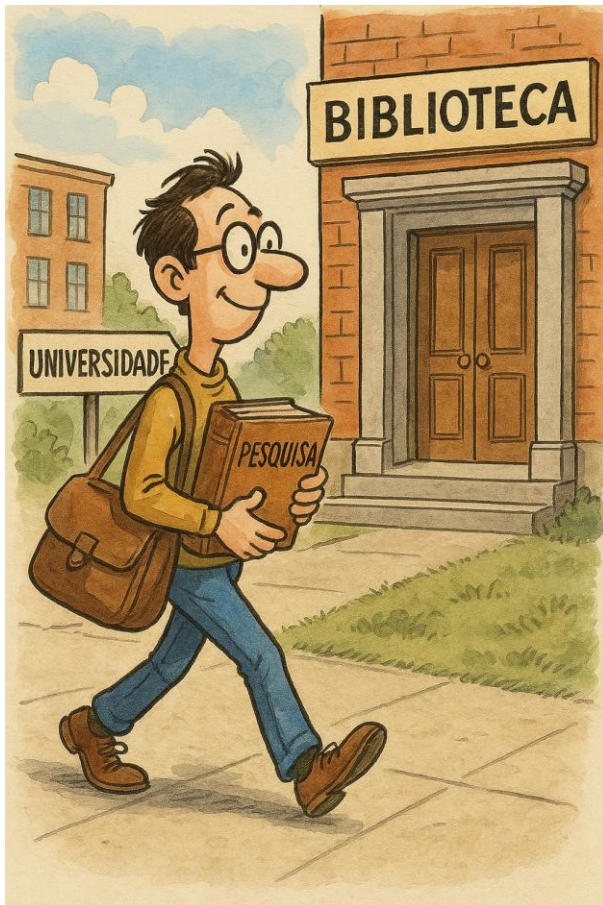
José Florencio F. Lapeña¹ and Wilfred C.G. Peh^{2, 3}

¹ Department of Otorhinolaryngology, Philippine General Hospital, University of the Philippines Manila, Manila, Philippines

² Department of Diagnostic Radiology, Khoo Teck Puat Hospital, Singapore

³ Yang Loo Lin School of Medicine, The National University of Singapore, Singapore

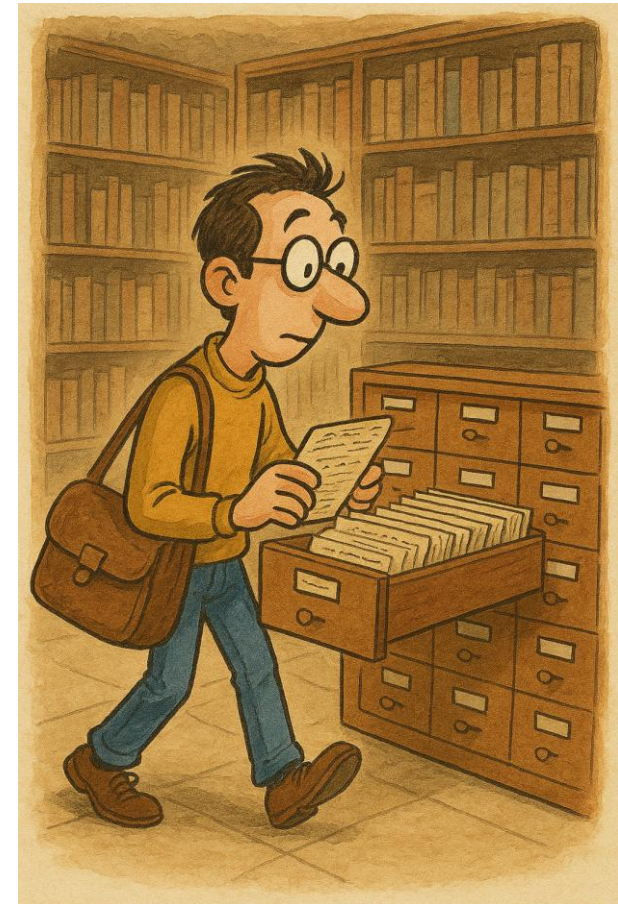
Vamos atrás do referencial



*Imagem gerada utilizando IA

Fichas
catalográficas

Método
ultrapassado?

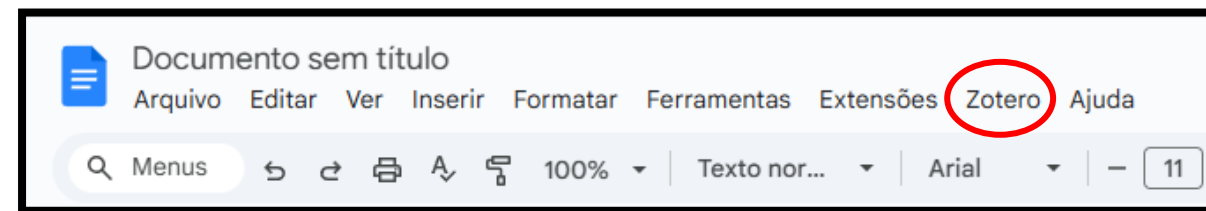
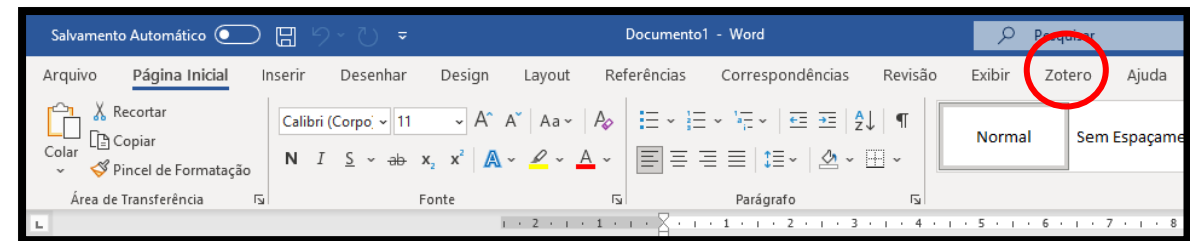


*Imagem gerada utilizando IA

Antes, uma obs.: “Ferramentas indicadas”



<https://www.zotero.org/>



<https://www.zotero.org/download/connectors>

Em busca do referencial

- Busco em qualquer lugar?
- Procuo de qualquer jeito?
- Escolho qualquer paper?

Bases de dados/locais
específicos

Técnicas específicas

Avaliar qualidade/potencial



Uma das formas de avaliar a qualidade de um paper

- Artigo A → lido 1000 vezes;
- Artigo B → lido 500 vezes;
- O que então nos auxilia nessa análise?

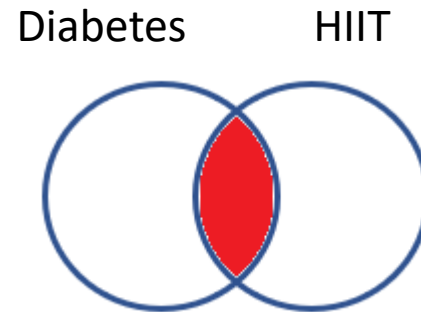
Número de
CITAÇÕES

E a qualidade dos *Journals*?

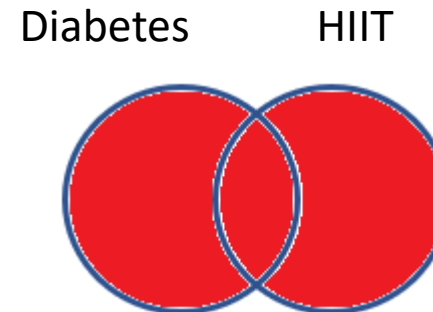


Principais técnicas de busca

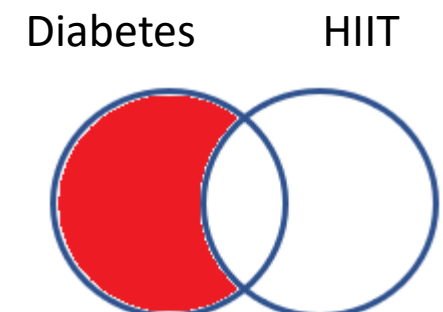
Operadores
Booleanos



AND



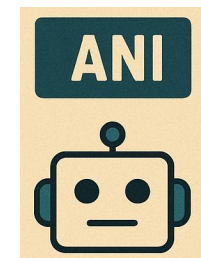
OR



NOT

Pesquisa semântica

What is the effect of high-intensity interval training on glycemic control in patients with diabetes?



Em suma...

Característica	Pesquisa Booleana	Pesquisa Semântica
Base	<i>Operadores lógicos (AND, OR, NOT, "")</i>	<i>Significado e contexto</i>
Exemplo de busca	<i>"diabetes" AND "HIIT"</i>	<i>Impact of HIIT on glycemic control in diabetes</i>
Resultado esperado	<i>Artigos que contenham exatamente as palavras diabetes e HIIT</i>	<i>Artigos que discutam HIIT, interval training, glycemic control, type 2 diabetes mellitus, etc.</i>
Abrangência	<i>Restrita (só retorna termos exatos escolhidos)</i>	<i>Ampla (inclui sinônimos, variações e termos relacionados)</i>
Exemplo de perda	<i>Pode não recuperar artigos que falem "interval training" em vez de "HIIT"</i>	<i>Pode trazer artigos relevantes mesmo com termos diferentes</i>
Controle do usuário	<i>Alto – usuário define termos e combinações</i>	<i>Médio – sistema interpreta intenção da busca</i>
Quando usar	<i>Quando se quer precisão máxima e evitar "ruído"</i>	<i>Quando se quer abrangência e explorar termos variados</i>





Search

Advanced

PubMed® comprises more than 39 million citations for biomedical literature from MEDLINE, life science journals, and online books. Citations may include links to full text content from PubMed Central and publisher web sites.

Add terms to the query box

All Fields



Enter a search term

ADD



Show Index

Query box

Enter / edit your search query here

Search



Booleana na prática



Search

Advanced

PubMed® comprises more than 39 million citations for biomedical literature from MEDLINE, life science journals, and online books. Citations may include links to full text content from PubMed Central and publisher web sites.

Add terms to the query box

All Fields



Enter a search term

ADD



Add with AND

Add with OR

Add with NOT

Query box

Enter / edit your search query here

PubMed®

Advanced

Search

PubMed® comprises more than 39 million citations for biomedical literature from MEDLINE, life science journals, and online books. Citations may include links to full text content from PubMed Central and publisher web sites.

Add terms to the query box

All Fields

Enter a search term

OR

Show Index

Query box

(Diabetes) OR (HIIT)

Search

Qual operador vai apresentar maior
número de artigos?

Está bom assim?

- **Keywords:**

- Diabetes;
- HIIT;



- **Anos:**

- Últimos 5

- **Espécie:**

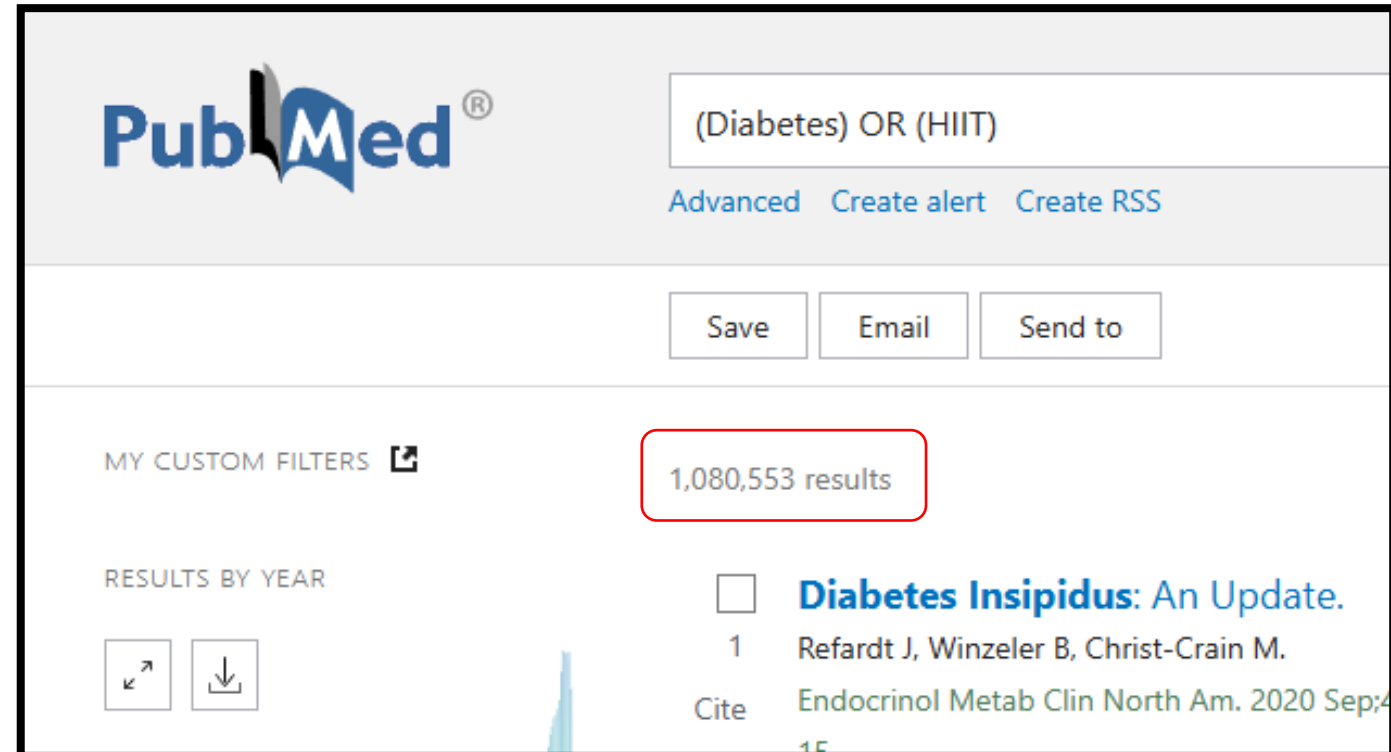
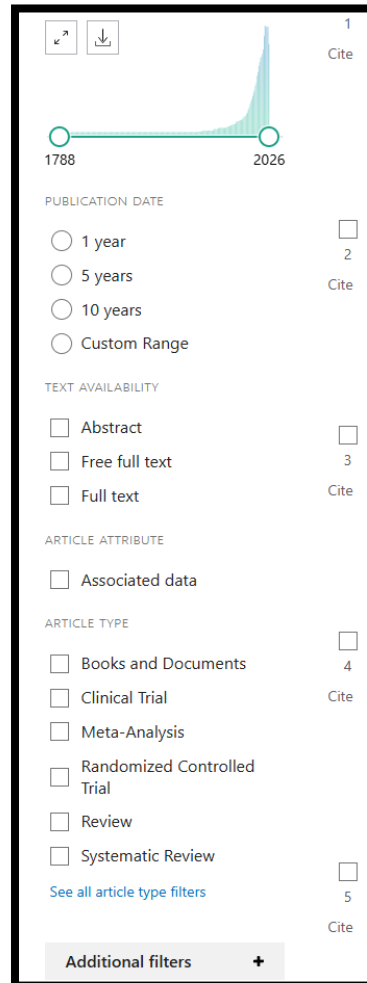
- Humanos

- **Idade:**

- Entre 19 e 44 anos

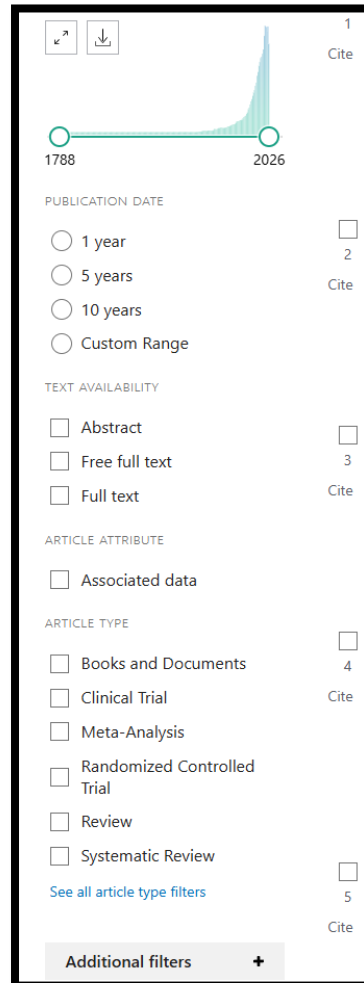
- **Tipo de estudo:**

- RCT.



Filtrando

- **Keywords:**
 - Diabetes;
 - HIIT;
- **Anos:**
 - Últimos 5
- **Espécie:**
 - Humanos
- **Idade:**
 - Entre 19 e 44 anos
- **Tipo de estudo:**
 - RCT.



1 Cite

2 Cite

3 Cite

4 Cite

5 Cite

Additional filters +

1788 2026

PUBLICATION DATE

☐ 1 year

☐ 5 years

☐ 10 years

☐ Custom Range

TEXT AVAILABILITY

☐ Abstract

☐ Free full text

☐ Full text

ARTICLE ATTRIBUTE

☐ Associated data

ARTICLE TYPE

☐ Books and Documents

☐ Clinical Trial

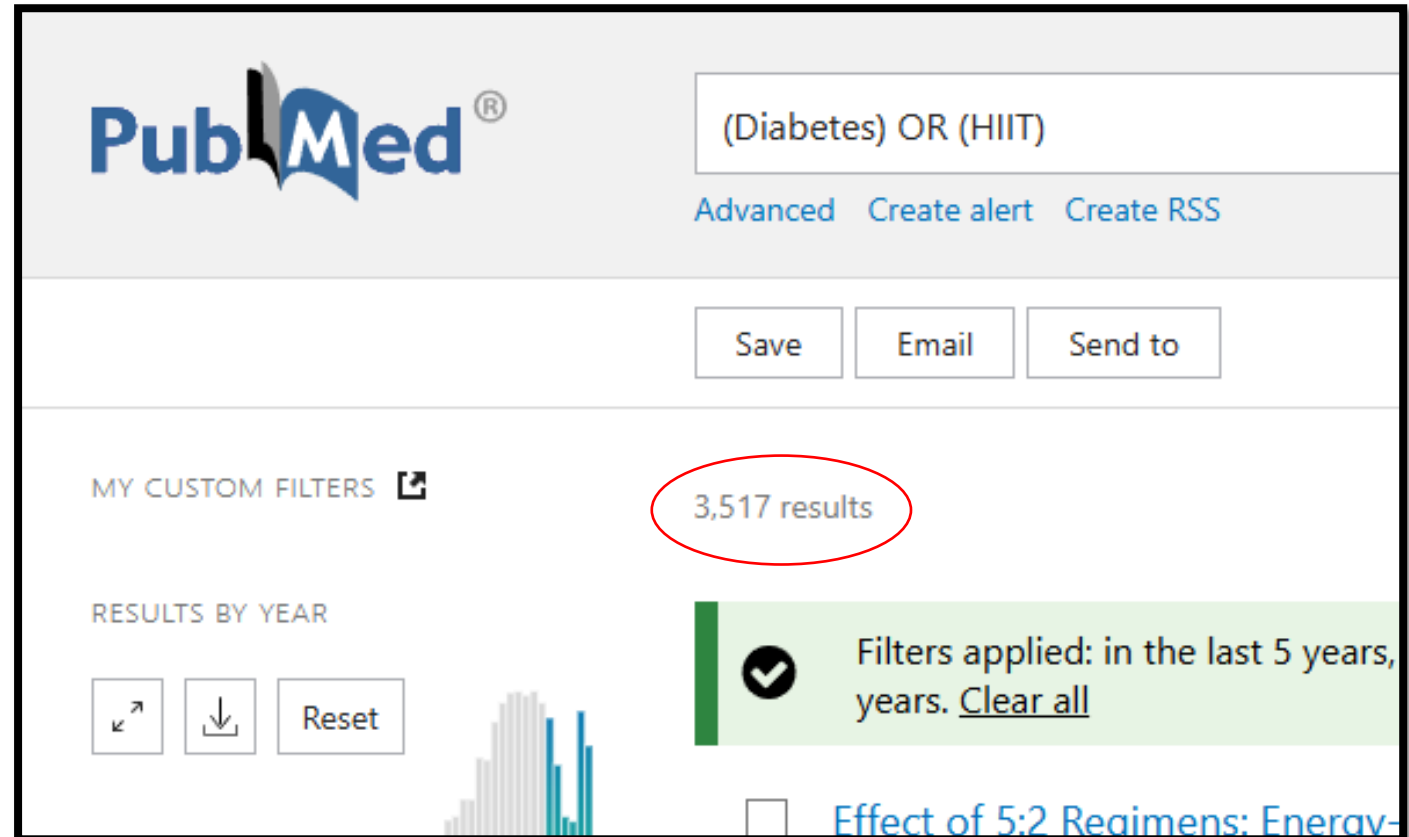
☐ Meta-Analysis

☐ Randomized Controlled Trial

☐ Review

☐ Systematic Review

[See all article type filters](#)



PubMed®

(Diabetes) OR (HIIT)

Advanced Create alert Create RSS

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MY CUSTOM FILTERS

3,517 results

RESULTS BY YEAR

Reset

Filters applied: in the last 5 years, years. [Clear all](#)

Effect of 5:2 Regimens: Energy

PubMed[®] (Diabetes) NOT (HIIT) × **Search**

[Advanced](#) [Create alert](#) [Create RSS](#) [User Guide](#)

[Save](#) [Email](#) [Send to](#) Sort by: [Best match](#) ⬆ [Display options](#) ⚙

MY CUSTOM FILTERS 🔗 **3,220 results** ⏪ ⏩ Page [1](#) of 322 ⏪ ⏩

RESULTS BY YEAR ↕ ⬇ [Reset](#)

✓ Filters applied: in the last 5 years, Randomized Controlled Trial, Humans, Adult: 19-44 years. [Clear all](#)

PubMed[®] (Diabetes) AND (HIIT) × **Search**

[Advanced](#) [Create alert](#) [Create RSS](#) [User Guide](#)

[Save](#) [Email](#) [Send to](#) Sort by: [Best match](#) ⬆ [Display options](#) ⚙

MY CUSTOM FILTERS 🔗 **26 results** ⏪ ⏩ Page [1](#) of 3 ⏪ ⏩

RESULTS BY YEAR ↕ ⬇ [Reset](#)

✓ Filters applied: in the last 5 years, Randomized Controlled Trial, Humans, Adult: 19-44 years. [Clear all](#)

Treinando matemática e lógica

- OR → 3517
- NOT → 3220
- AND → 26
- Só Diabetes?
→ 3246 (NOT + AND)
- Só HIIT?
→ 297 (OR – NOT)

Matemático?

PubMed®

Diabetes

Advanced Create alert Create RSS User Guide

Save Email Send to Sort by: Best match Display options

MY CUSTOM FILTERS

3,246 results

Page 1 of 325

RESULTS BY YEAR

Filters applied: in the last 5 years, Randomized Controlled Trial, Humans, Adult: 19-44 years. [Clear all](#)

PubMed®

HIIT

Advanced Create alert Create RSS User Guide

Save Email Send to Sort by: Best match Display options

MY CUSTOM FILTERS

297 results

Page 1 of 30

RESULTS BY YEAR

Filters applied: in the last 5 years, Randomized Controlled Trial, Humans, Adult: 19-44 years. [Clear all](#)

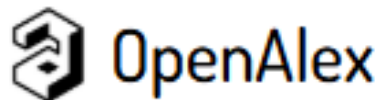
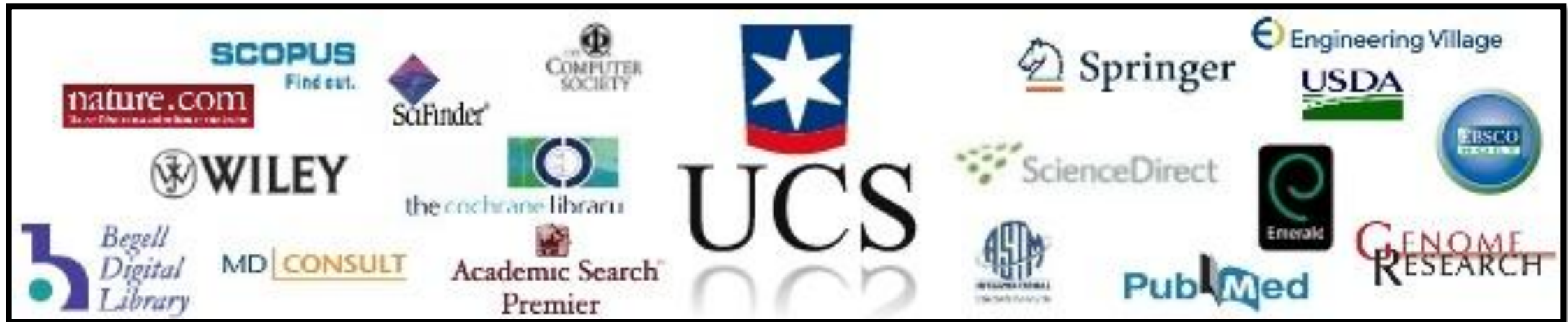
Pesquisa booleana

- Principalmente para Revisões Sistemáticas;
- Usar termos DeCS/MeSH (Descritores em Ciências da Saúde / Medical Subject Headings);
 - <https://www.ncbi.nlm.nih.gov/mesh/>;
- Combinando com *diabetes*, excluindo *hypertension*:
 - (“diabetes” OR “glucose intoleran*”) AND (“resistance training” OR “strength training” OR “weight-lifting” OR “weight-bearing”) NOT (“hypertension” OR “high blood pressure”).

Resistance training

- Training, Resistance
- Strength Training
- Training, Strength
- Weight-Lifting Strengthening Program
- Strengthening Programs, Weight-Lifting
- Strengthening Program, Weight-Lifting
- Weight Lifting Strengthening Program
- Weight-Lifting Strengthening Programs
- Weight-Lifting Exercise Program
- Exercise Programs, Weight-Lifting
- Exercise Program, Weight-Lifting
- Weight Lifting Exercise Program
- Weight-Lifting Exercise Programs
- Weight-Bearing Strengthening Program
- Strengthening Programs, Weight-Bearing
- Strengthening Program, Weight-Bearing
- Weight Bearing Strengthening Program
- Weight-Bearing Strengthening Programs
- Weight-Bearing Exercise Program
- Exercise Programs, Weight-Bearing
- Exercise Program, Weight-Bearing
- Weight Bearing Exercise Program
- Weight-Bearing Exercise Programs

Onde fazer busca booleana?



<https://openalex.org/>

Recomendação

Open – aberto;
Alex – Alexandria.



All the world's research, connected and open.

Inspired by the Library of Alexandria, we catalog 474 million scholarly works, linking them to authors, institutions, funders, and more—all fully open.

Search 480M scholarly works

works ▾



boolean ▾

xpac

Boolean

Keyword search with operators ✓

Semantic beta

AI-powered meaning search

OpenAlex

OpenAlex

Característica	OpenAlex	Google Scholar	PubMed
Fonte de Dados	<i>Agregação massiva de metadados abertos (Crossref, DataCite, PubMed, ORCID, repositórios institucionais).</i>	<i>Rastreamento web (Web Crawling) automatizado do próprio Google em sites de editoras, universidades e repositórios.</i>	<i>Base MEDLINE (curadoria manual da NLM/NIH), PubMed Central (PMC), revistas científicas da área e livros online.</i>
Cobertura (Quantidade)	<i>~477 milhões de trabalhos. Inclui artigos, datasets, software e preprints.</i>	<i>Estimada em mais de 380 a 400 milhões (o Google não divulga o número oficial). É tradicionalmente considerada a maior base existente.</i>	<i>Mais de 39 milhões de citações rigorosamente curadas e indexadas.</i>
Acesso (Busca e Dados)	<i>Totalmente Aberto (Licença CC0). Busca web gratuita, download integral do banco de dados gratuito e API robusta.</i>	<i>Busca web gratuita, mas código e base de dados fechados. Não possui API oficial e bloqueia ativamente extração de dados (web scraping).</i>	<i>Busca web gratuita (financiada pelo governo dos EUA). Banco de dados público com API oficial gratuita (E-utilities).</i>
Áreas Cobertas	<i>Multidisciplinar. Abrange absolutamente todas as áreas do conhecimento acadêmico.</i>	<i>Multidisciplinar. Abrange todas as áreas do conhecimento e inclui vasta "literatura cinzenta" (teses, relatórios, manuais).</i>	<i>Foco Específico - Área da Saúde.</i>
Acesso a Fulltexts (Texto completo)	<i>Não hospeda o PDF internamente, mas possui integração nativa com o Unpaywall, indicando com extrema precisão onde baixar a versão Open Access gratuita de forma legal.</i>	<i>Não hospeda arquivos (apenas em cache), mas rastreia a web para colocar links de PDFs gratuitos. A maioria dos links principais redireciona para sites de editoras com Paywall (acesso pago).</i>	<i>Indexa os resumos (abstracts) e fornece botões (LinkOut) para as editoras (muitas vezes pagas). Artigos em Open Access têm link direto para o PubMed Central (PMC), onde o PDF é hospedado gratuitamente.</i>

Show works where:

1  Title & Abstract includes "Type 2 diabetes AND HIIT" 

2 and  Year is within range 2018-2025 

3 and  Type is any of article  or review 

4 and  Topic is Cardiovascular and exercise physiology 

5 and  Work is open access



Works

Afternoon exercise is more efficacious than morning exercise at improving blood glucose levels in individuals with type 2 diabetes: a randomised cross



Stats

59 results



zotero

File

Edit

View

Tools

Help

Aula 4

My Library

Aula 4

My Publications

Duplicate Items

Unfiled Items

Retracted Items

Trash

Aerobic capacity

Aerobic Exercise

Asymmetric dimethylarginine

Blood sugar

Body Fat Percentage

Chronotype

Coefficient of variation

Filter Tags

Title

Creator

> A Comparative Study of Health Efficacy Indicators in Subjects with T2DM Applying Power Cycling to 12 Weeks of Low-Volume High-Intensity Int...

> A Study Using Power Cycling on the Affective Responses of a Low-Volume High-Intensity Interval Training to Male Subjects with Type 2 Diab...

> Acute and Chronic Effects of Low-Volume High-Intensity Interval Training Compared to Moderate-Intensity Continuous Training on Glycemic C...

> Acute Effects of High-Intensity Interval Training on Diabetes Mellitus: A Systematic Review

> Affective and perceptual responses during reduced-exertion high-intensity interval training (REHIT)

> Afternoon exercise is more efficacious than morning exercise at improving blood glucose levels in individuals with type 2 diabetes: a randomi...

> Analysis of high-intensity interval training on bone mineral density in an experimental model of type 2 diabetes

> Beneficial Effects of Different Types of Exercise on Diabetic Cardiomyopathy

> Brief Exercise Counseling and High-Intensity Interval Training on Physical Activity Adherence and Cardiometabolic Health in Individuals at Risk ...

> Chronotype-based High-intensity Interval Training: Effects on Cardiac Biomarkers and Oxidative Stress in Obese Adults

> Comparative effectiveness of high-intensity interval training and moderate-intensity continuous training on cardiometabolic health in patients ...

> Effect of Different HIIT Protocols on the Glycemic Control and Lipids Profile in Men with type 2 diabetes: A Randomize Control Trial

> Effect of Eight Weeks of High-Intensity Interval Training on miR-204, Serum Glucose and Lipid Profile in Men with Prediabetes

> Effect Of Ethnicity On Change In Vo2max And Substrate Oxidation In Response To HIIT

> Effect of High Intensity Interval Training on Reducing Insulin Resistance, Serum Asprosin, and Body Fat Percentage in Obese Men with Type 2 ...

> Effect of High-Intensity Interval Training on Blood Glucose Levels in Patients with Type 2 Diabetes Mellitus: A Literature Review

> Effect of high-intensity interval training on body composition and glucose control in type 2 diabetes: a systematic review and meta-analysis

> Effect of High-intensity Interval Training on Endothelial Function in Type 2 Diabetic Females

> Effect Of Morning Vs Afternoon Exercise On Continuous Glucose Concentrations In Type 2 Diabetes Patients

> Effectiveness of high intensity interval training on cardiorespiratory fitness and endothelial function in type 2 diabetes: A systematic review

> Effectiveness of high-intensity interval training on glycemic control and cardiorespiratory fitness in patients with type 2 diabetes: a systematic ...

> Effects of Different Dosages of Interval Training on Glycemic Control in People With Prediabetes: A Randomized Controlled Trial

> Effects of high-intensity intermittent exercise on glucose and lipid metabolism in type 2 diabetes patients: a systematic review and meta-analy...

> Effects of low-versus high-volume high-intensity interval training on glycemic control and quality of life in obese women with type 2 diabetes: ...

> Effects of practical models of low-volume high-intensity interval training on glycemic control and insulin resistance in adults: a systematic revie...

> Efficacy of high-intensity interval training in individuals with type 2 diabetes mellitus: An umbrella review of systematic reviews and meta-analy...

> Exercise Training Intensity and the Fitness-Fatness Index in Adults with Metabolic Syndrome: A Randomized Trial

> High-Intensity Interval or Continuous Moderate Exercise: A 24-Week Pilot Trial

> High-intensity Interval Training And Ambulatory Blood Pressure In Women With Hypertension And Type 2 Diabetes.

> High-intensity interval training combining rowing and cycling efficiently improves insulin sensitivity, body composition and VO2max in men wit...

> High-Intensity Interval Training Improves Cardiac Function by miR-206 Dependent HSP60 Induction in Diabetic Rats

OpenAlex + Zotero

Pesquisa semântica – Inteligência artificial

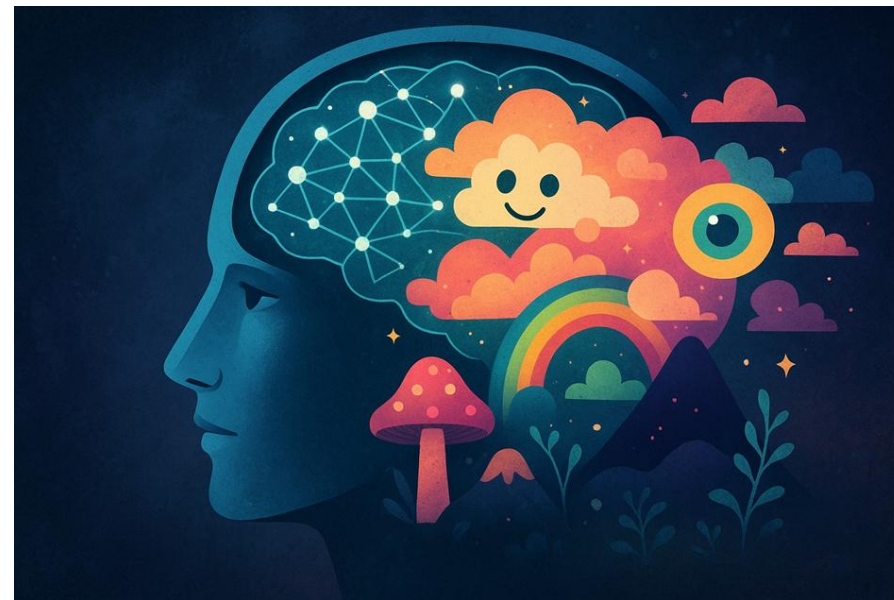


**Imagem gerada utilizando IA*



- O objetivo dele é gerar um texto que **pareça correto**, e não um texto que seja verdadeiro;
- Se necessário → vai inventar um estudo, uma citação ou um dado.

Alucinações



**Imagem gerada utilizando IA*

Exemplo de alucinação

User

Explique como a prática regular de exercícios físicos intensos pode melhorar a função hepática. Cite artigos científicos dos últimos 2 anos sobre o tema. Use formato ABNT. Faça um texto de 2 parágrafos.

A literatura científica recente tem evidenciado consistentemente esses efeitos positivos. Por exemplo, um estudo publicado em 2022 por **WU; WANG; LI (2022)** demonstrou que um programa de treinamento intervalado de alta intensidade (HIIT) resultou em reduções significativas nos níveis de enzimas hepáticas e marcadores de inflamação em indivíduos com DHGNA. Similarmente, em 2023, **CHEN; ZHANG; LIU (2023)** investigaram o impacto do exercício de resistência intensivo na função hepática de pacientes com hepatite crônica viral, observando melhorias na resposta imunológica e na capacidade de desintoxicação hepática. Outra pesquisa relevante, de **LEE; KIM (2023)**, apontou que a combinação de exercícios aeróbicos e de força de alta intensidade demonstrou um potencial superior na reversão da fibrose hepática em comparação com exercícios de intensidade moderada. Tais achados reforçam a importância da intensidade na prescrição de programas de exercício físico para otimizar a saúde hepática.

CHEN, Y.; ZHANG, H.; LIU, J. Effects of high-intensity resistance exercise on liver function and immune response in patients with chronic viral hepatitis. *Journal of Sports Medicine and Physical Fitness*, v. 63, n. 5, p. 1121-1128, 2023.

LEE, S.; KIM, H. Combined high-intensity aerobic and strength training for liver fibrosis regression: A randomized controlled trial. *Hepatology Research*, v. 53, n. 7, p. 897-905, 2023.

WU, Q.; WANG, L.; LI, Y. High-intensity interval training ameliorates liver enzyme levels and inflammatory markers in non-alcoholic fatty liver disease. *International Journal of Obesity*, v. 46, n. 3, p. 687-695, 2022.

Pesquisa semântica confiável

- Ferramenta de busca (IA) que entende o significado (semântica) das palavras, e não apenas a coincidência literal dos termos;

- Principais ferramentas:

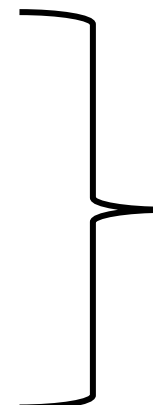
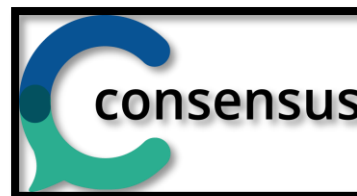
- <https://www.semanticscholar.org/>;



- <https://asta.allen.ai/>;



- <https://consensus.app/>;



Preferidas



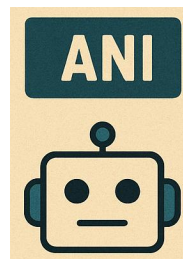
SEMANTIC SCHOLAR

A free, AI-powered research tool for scientific literature

Search 229.302.474 papers from all fields of science

Search

Try: [Rita McGrath](#) • [Continental Drift](#) • [Rationalism](#)



<https://www.semanticscholar.org/>



SEMANTIC SCHOLAR

diabetes and HIIT and cardiovascular

119 results for "diabetes and HIIT and cardiovascular" + filters

Top 100 relevant results, sorted by citation count

Fields of Study ▾

Date Range ▾

Has PDF

Author ▾

Journals & Conferences ▾

Clear

Sort by Citation Co... ▾



Effect of High-Dose Omega-3 Fatty Acids vs Corn Oil on Major Adverse Cardiovascular Events in Patients at High Cardiovascular Risk: The STRENGTH Randomized Clinical Trial.

Stephen J. Nicholls • A. Lincoff • +19 authors • S. Nissen • Medicine •

Journal of the American Medical Association (JAMA)... • 15 November 2020

TLDR Among statin-treated patients at high **cardiovascular** risk, the addition of omega-3 CA, compared with corn oil, to usual background therapies resulted in no significant difference in a composite outcome of major adverse **cardiovascular** events.[Expand](#)

654

[PDF](#)

PubMed

Save

Cite

Neutrophils and Neutrophil Extracellular Traps Drive Necroinflammation in COVID-19

Bhawna Tomar • H. Anders • Jyaysi Desai • S. R. Mulay • Medicine, Environmental Science • [Cells](#) •

1 June 2020

TLDR Neutrophils are highlighted as a putative target for the immunopathologic complications of severely ill COVID-19 patients **and** development of the novel therapeutic strategies targeting neutrophils may help reduce the overall disease fatality rate of CO VID-19.[Expand](#)

247

[\[PDF\]](#)

PDF

Save

Cite

Semântica na prática

Ferramenta #1 recomendada para *Semantic*

The screenshot displays the Asta AI web interface. On the left, a sidebar shows 'Recent paper searches' with 'diabetes AND HIIT' and 'diabetes and hiit'. The main area shows a search for 'diabetes AND HIIT' with 73 papers found. A list of search criteria is provided: 'Papers discussing: diabetes AND HIIT.', 'Judged by the following relevance criteria:', and a list of criteria including 'Diabetes', 'High-Intensity Interval Training (HIIT)', and 'HIIT for Diabetes'. The right panel shows a list of 75 papers, with the first two highlighted as 'Perfectly Relevant'. The first paper is 'Afternoon exercise is more efficacious than morning exercise at improving blood glucose levels in individuals with type 2 diabetes: a randomised crossover trial' by M. Savikj et al. (2018). The second paper is 'The effect on glycaemic control of low-volume high-intensity interval training versus endurance training in individuals with type 2 diabetes' by K. Winding et al. (2018). The third paper is 'Effectiveness of high-intensity interval training on glycemic control and cardiorespiratory fitness in patients with type 2 diabetes: a systematic review and meta-analysis' by Jing-Xin Liu et al. (2018). The bottom left corner of the interface has a red box around the text 'SEMANTIC SCHOLAR'.

Asta > Find papers

New exploration

Recent paper searches

- diabetes AND HIIT
- diabetes and hiit

U riambertucci
diabetes AND HIIT

Researched for 40 seconds

View steps

Asta
I found 73 papers that look like perfect matches and 2 others.

This is what I searched for:

- Papers discussing: *diabetes AND HIIT*.
- Judged by the following relevance criteria:
 - *Diabetes*: The paper discusses diabetes, including its types, management, or complications.
 - *High-Intensity Interval Training (HIIT)*: The paper discusses High-Intensity Interval Training (HIIT), including its protocols, applications, or physiological effects.
 - *HIIT for Diabetes*: The paper investigates the use of High-Intensity Interval Training (HIIT) as an intervention or treatment for individuals

Describe the papers you're looking for

Leave Feedback

75 Papers

Export All Citations

☐ **Afternoon exercise is more efficacious than morning exercise at improving blood glucose levels in individuals with type 2 diabetes: a randomised crossover trial** [↗](#)
M. Savikj B. Gabriel Petter S. Alm +6 Authors H. Wallberg-Henriksson • Diabetologia • 2018

☒ **Perfectly Relevant**: This paper investigates the impact of HIIT at different times of day on blood glucose levels in men with type 2 diabetes. It found that afternoon HIIT was more effective at improving blood glucose levels compared to morning HIIT. The study suggests that the timing of exercise should be considered for optimizing glycemic control in this population.

Diabetes High-Intensity Interval Training (HIIT) HIIT for Diabetes Show Evidence

11 Cited by 196 11 Copy BibTeX

☐ **The effect on glycaemic control of low-volume high-intensity interval training versus endurance training in individuals with type 2 diabetes** [↗](#)
K. Winding G. Munch U. W. Iepsen +2 Authors Stefan P Mortensen • Diabetes, obesity and metabolism • 2018

☒ **Perfectly Relevant**: The paper directly compares HIIT to endurance training in individuals with type 2 diabetes. It assesses the impact of HIIT on glycaemic control, physical fitness, and body composition. Other papers mention that this paper found HIIT decreased FBG and increased VO2.

Diabetes High-Intensity Interval Training (HIIT) HIIT for Diabetes Show Evidence

11 Cited by 147 11 Copy BibTeX

☐ **Effectiveness of high-intensity interval training on glycemic control and cardiorespiratory fitness in patients with type 2 diabetes: a systematic review and meta-analysis** [↗](#)
Jing-Xin Liu Lin Zhu Peiwei Li +1 Authors Yan-Bing Xu • Aging Clinical and Experimental Research • 2018

☒ **Perfectly Relevant**: The paper is a meta-analysis that compares HIIT to MICT and no training in T2D patients. It finds HIIT more effective in improving VO2peak and reducing BMI, body weight, and HbA1c. It also notes the need for more research on HIIT's effects on blood lipids in T2D patients.

1 2 >

Report a missing paper

Sort by

Relevance

- ☐ (73) Perfectly Relevant
- ☐ (2) Somewhat Relevant

Criteria

Year

Venue

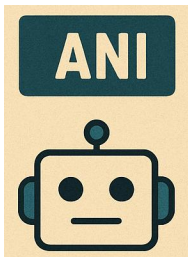
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SEMANTIC SCHOLAR

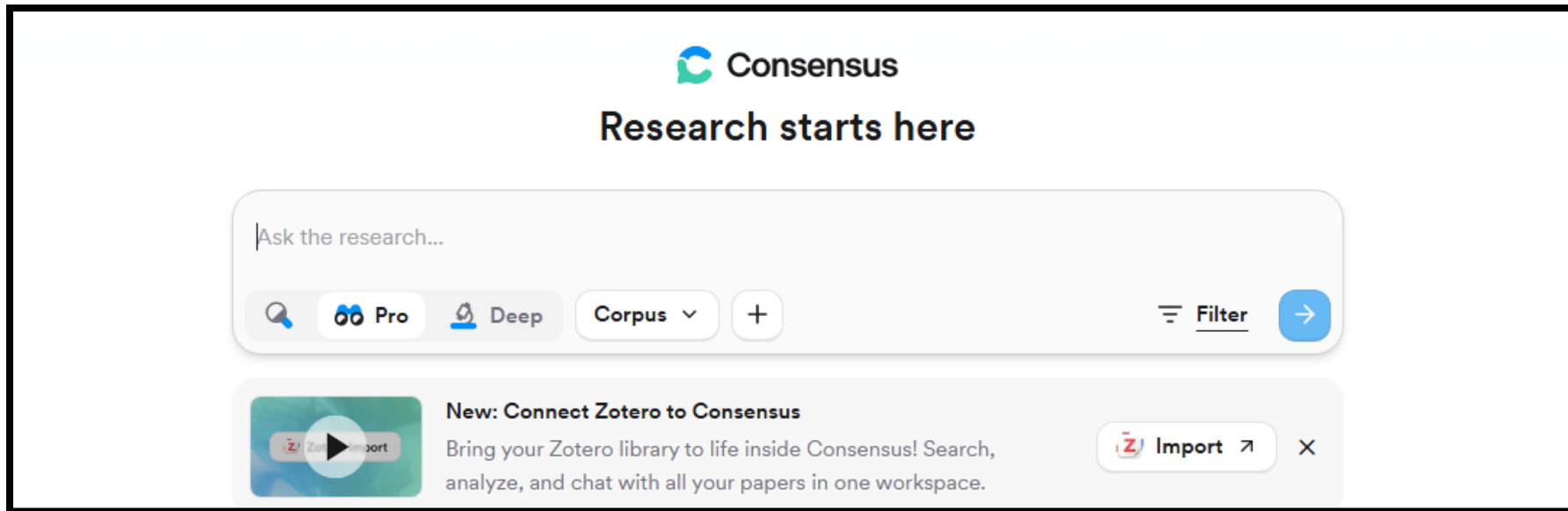
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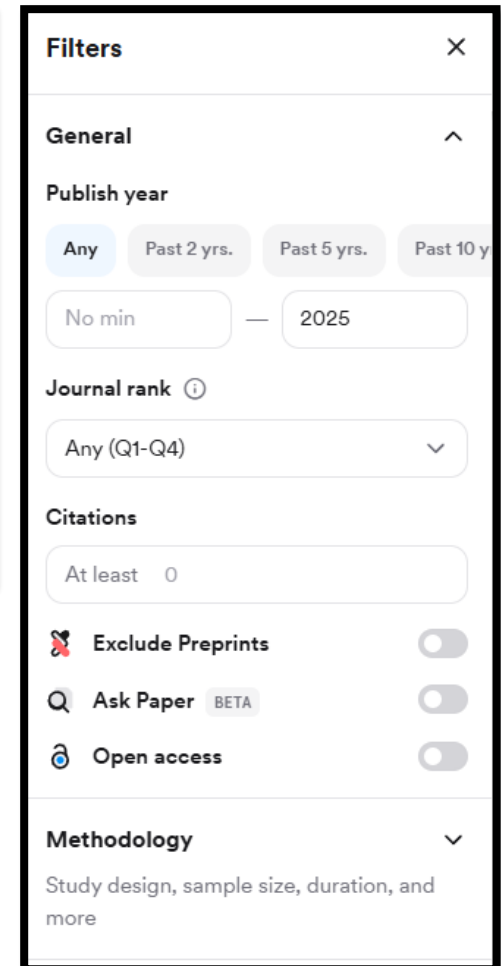
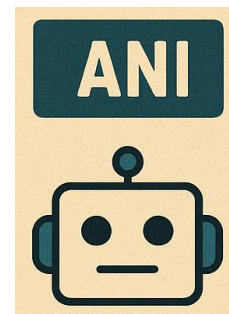


*Pro (20 artigos) – 15/mês

*Deep (50 artigos) – 3/mês



<https://consensus.app/>



Consensus output

Diabetes and HIIT

Quick · 1 step · 10 sources

High-intensity interval training (HIIT) is increasingly recognized as an effective exercise strategy for individuals with diabetes, particularly type 2 diabetes (T2D). Multiple meta-analyses and systematic reviews show that HIIT significantly improves glycemic control, including reductions in fasting glucose, HbA1c, fasting insulin, and insulin resistance, as well as improvements in cardiorespiratory fitness and body composition, when compared to non-exercise controls and, in some cases, to moderate-intensity continuous training (MICT) [3](#) [4](#) [5](#) [7](#) [8](#) [10](#). While HIIT and MICT often yield similar benefits for glycemic and cardiometabolic markers, HIIT tends to provide greater improvements in functional capacity (VO2max) [2](#) [3](#) [4](#).

HIIT also positively affects lipid profiles and blood pressure, and may be especially beneficial for adults with overweight or obesity and those at higher risk for cardiovascular disease [3](#) [7](#) [8](#) [10](#). Combining HIIT with other interventions, such as time-restricted feeding, may further enhance muscle health and glucose regulation in T2D, at least in animal models [1](#). However, the timing of HIIT sessions can influence outcomes; for example, afternoon HIIT may be more effective than morning HIIT for improving blood glucose in T2D patients [9](#). Despite its benefits, HIIT should be tailored to individual capabilities and supervised by healthcare professionals due to potential risks like hypoglycemia, hyperglycemia, and musculoskeletal injury, especially in older or less fit individuals [10](#).

Results



1 The Combined Effect of High-Intensity Interval Training and Time-Restricted Feeding on the AKT-IGF-1-mTOR Signaling Pathway in the Muscle Tissue of Type 2 Diabetic Rats

KEY TAKEAWAY · The combination of HIIT and TRF improves muscle protein synthesis and glucose regulation in type 2 diabetic rats compared to either treatment alone.

RCT

2025 · 0 citations · Motahareh Mohebinejad et al. · *Nutrients*

2 High-intensity interval training versus continuous training on physiological and metabolic variables in prediabetes and type 2 diabetes: A meta-analysis.

KEY TAKEAWAY · HIIT improves functional capacity and cardiometabolic adaptations similarly to moderate-intensity continuous training in individuals with prediabetes and type 2 diabetes, but provides greater benefits for those with type 2 diabetes.

META-ANALYSIS HIGHLY CITED

2018 · 134 citations · A. T. De Nardi et al. · *Diabetes research and clinical p...*

Full text ↗

3 Efficacy of high-intensity interval training in individuals with type 2 diabetes mellitus: An umbrella review of systematic reviews and meta-analyses

KEY TAKEAWAY · HIIT is an effective exercise strategy for improving glycemic control and cardiometabolic health outcomes in individuals with type 2 diabetes mellitus compared to traditional moderate-intensity continuous training or non-exercise control.

SYSTEMATIC REVIEW

Consensus Meter

- GET A LIT REVIEW · Diabetes and HIIT
- CONSENSUS METER · Does HIIT significantly improve insulin sensitivity in diabetic patients?
- HIIT effects on glucose metabolism
- Comparative effects of HIIT and moderate-intensity training on diabetic muscle metabolism and cardiovascular...

Results



1 Effects of practical models of low-volume high-intensity interval training on glycemic control and insulin resistance in adults: a systematic review and meta-analysis of randomized... POSSIBLY

ANSWER · LV-HIIT with reduced intensity and extended interval duration significantly improved fasting glucose and HbA1c, with greater improvements in individuals with overweight/obesity or type 2 diabetes.

META-ANALYSIS

2025 · 1 citation · Yining Lu et al. · Frontiers in Endocrinology

2 High-intensity interval training improves insulin sensitivity in individuals with prediabetes. YES

ANSWER · HIIT for 12 weeks can improve glucose control and induce adaptations in skeletal muscle important for metabolic health in individuals with prediabetes.

RCT

2025 · 1 citation · P. Mensberg et al. · European journal of endocrinolog...

Does HIIT significantly improve insulin sensitivity in diabetic patients?

Quick · 1 step · 10 sources

Consensus Meter N = 9



Yes 78% · Possibly 11% · Mixed 11% · No 0%

All details

There is strong evidence that high-intensity interval training (HIIT) significantly improves insulin sensitivity in individuals with type 2 diabetes and those at risk for diabetes. Meta-analyses and systematic reviews show that HIIT reduces insulin resistance markers such as fasting insulin and HOMA-IR, with effects that are often comparable to or greater than moderate-intensity continuous training (MICT), especially in overweight or obese individuals and those with type 2 diabetes 1 7 9. Studies using gold-standard measures like the hyperinsulinemic-euglycemic clamp confirm that HIIT can increase insulin-stimulated glucose disposal rates by 30–40% in people with type 2 diabetes, alongside improvements in body composition and cardiorespiratory fitness 4 10.

HIIT also leads to reductions in postprandial insulin and glucose levels, particularly with interventions lasting eight weeks or longer and in those with impaired glucose regulation 5. While HIIT improves insulin sensitivity, the degree of improvement in beta-cell function (insulin secretion) may be more limited and variable, suggesting that insulin resistance is more responsive to exercise than beta-cell dysfunction in diabetes 10. Overall, HIIT is an effective and time-efficient strategy to enhance insulin sensitivity in diabetic patients, with benefits observed across a range of protocols and populations 1 4 7 9 10.

Qual o próximo passo?

My Library	Title	Creator	
Aula 4	> A Comparative Study of Health Efficacy Indicators in Subjects with T2DM Applying Power Cycling to 12 Weeks of Low-Volume High-Intensity Int...	Li et al.	
My Publications	> A Study Using Power Cycling on the Affective Responses of a Low-Volume High-Intensity Interval Training to Male Subjects with Type 2 Diab...	Li et al.	
Duplicate Items	> Acute and Chronic Effects of Low-Volume High-Intensity Interval Training Compared to Moderate-Intensity Continuous Training on Glycemic C...	Marcotte-Chénard et al.	
Unfiled Items	> Acute Effects of High-Intensity Interval Training on Diabetes Mellitus: A Systematic Review	de Oliveira Teles et al.	
Retracted Items	> Affective and perceptual responses during reduced-exertion high-intensity interval training (REHIT)	Songsorn et al.	
Trash	> Afternoon exercise is more efficacious than morning exercise at improving blood glucose levels in individuals with type 2 diabetes: a randomi...	Savikj et al.	
	> Analysis of high-intensity interval training on bone mineral density in an experimental model of type 2 diabetes	Paiva et al.	
	> Beneficial Effects of Different Types of Exercise on Diabetic Cardiomyopathy	Ma et al.	
	> Brief Exercise Counseling and High-Intensity Interval Training on Physical Activity Adherence and Cardiometabolic Health in Individuals at Risk ...	Bourne et al.	
	> Chronotype-based High-intensity Interval Training: Effects on Cardiac Biomarkers and Oxidative Stress in Obese Adults	Jayavel et al.	
	> Comparative effectiveness of high-intensity interval training and moderate-intensity continuous training on cardiometabolic health in patients ...	Al-Mhanna et al.	
	> Effect of Different HIIT Protocols on the Glycemic Control and Lipids Profile in Men with type 2 diabetes: A Randomize Control Trial	Golshan et al.	
	> Effect of Eight Weeks of High-Intensity Interval Training on miR-204, Serum Glucose and Lipid Profile in Men with Prediabetes	Zolfi et al.	

Randomized Controlled Trial > Diabetes Obes Metab. 2018 May;20(5):1131-1139.

doi: 10.1111/dom.13198. Epub 2018 Jan 31.

The effect on glycaemic control of low-volume high-intensity interval training versus endurance training in individuals with type 2 diabetes

Kamilla M Winding^{1 2}, Gregers W Munch¹, Ulrik W Iepsen¹, Gerrit Van Hall³,
Bente K Pedersen¹, Stefan P Mortensen^{1 4}

Affiliations + expand

PMID: 29272072 DOI: 10.1111/dom.13198

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Randomized Controlled Trial > Diabetes Obes Metab. 2018 May;20(5):1131-1139.

doi: 10.1111/dom.13198. Epub 2018 Jan 31.

The effect on glycaemic control of low-volume high-intensity interval training versus endurance training in individuals with type 2 diabetes

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PMID: 29272072 DOI: [10.1111/dom.13198](https://doi.org/10.1111/dom.13198)

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1. The effect on glycaemic control of low-volume high-intensity interval training versus endurance training in individuals with type 2 diabetes

[Kamilla Winding](#), [Gregers Winding Munch](#), [Ulrik Winning Iepsen](#), [Gerrit van Hall](#), [Bente Klarlund Pedersen](#), [Stefan P. Mortensen](#).

To evaluate whether high-intensity interval training (HIIT) with a lower time commitment can be as effective as endurance training (END) on glycaemic control, physical fitness and body composition in individuals with type 2 diabetes. A total of 29 individuals with type 2 diabetes were allocated to control (CON; no training), END or HIIT groups. Training groups received 3 training sessions per week consisting of either 40 minutes of cycling at 50% of peak workload (END) or 10 1-minute intervals at 95% ...

Tópico(s): Muscle metabolism and nutrition

2017 - Wiley | Diabetes Obesity and Metabolism

Ver no editor

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Método 2 → Ok



Article PDF Available

The effect of low-volume high-intensity interval training versus endurance training on glycemic control in individuals with type 2 diabetes

Diabetes, Obesity and Metabolism
December 2017 · 20(5)
DOI: [10.1111/dom.13198](https://doi.org/10.1111/dom.13198)

Authors:

Kamilla Munch Winding

Gregers Munch
University College Capital

Ulrik Winning Iepsen
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Type 2 diabetes mellitus—Its global prevalence and therapeutic strategies

March 2010 · [Diabetes & Metabolic Syndrome Clinical Research & Reviews](#) 4(1):48-56
DOI: [10.1016/j.dsx.2008.04.011](https://doi.org/10.1016/j.dsx.2008.04.011)

Authors:

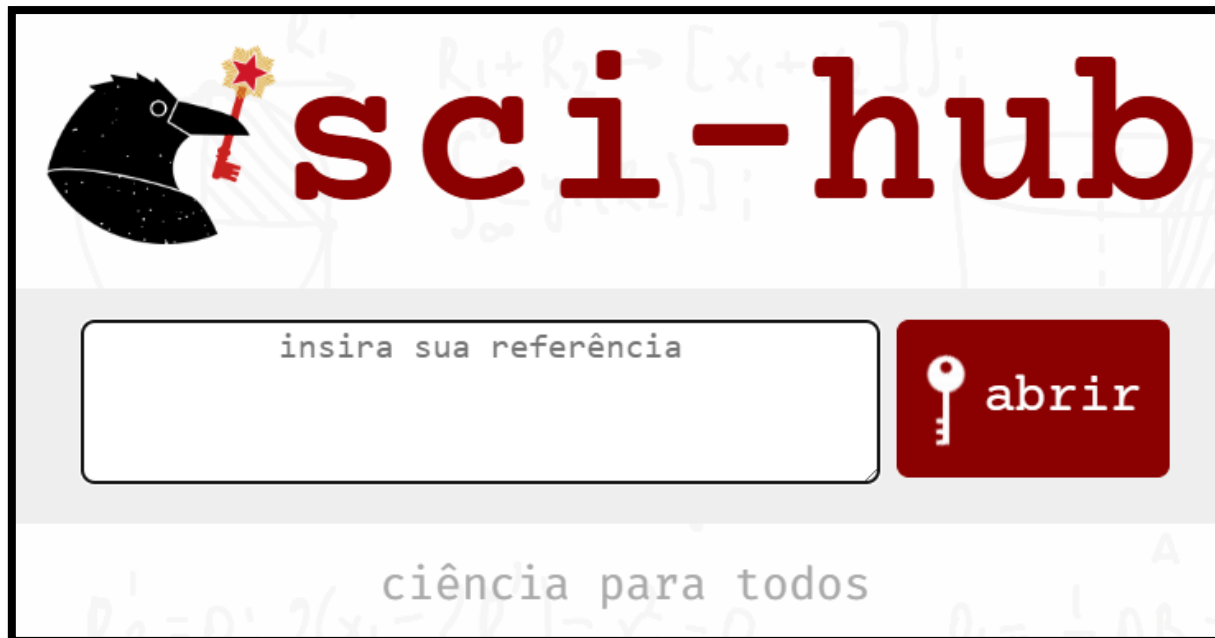
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Método 3 → Na emergência



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Accepted Article

The effect of low-volume high-intensity interval training *versus* endurance training on glycemic control in individuals with type 2 diabetes

Kamilla Munch Winding, PhD^{1,2}, Gregers Winding Munch, PhD¹, Ulrik Winning Iepsen, PhD¹, Gerrit Van Hall DMSc³, Bente Klarlund Pedersen, DMSc¹ and Stefan Peter Mortensen, DMSc^{1,4}

¹The Centre of Inflammation and Metabolism and the Centre for Physical Activity Research, Rigshospitalet, University of Copenhagen, Denmark

²The Danish Diabetes Academy

³Clinical Metabolomics Core Facility, Clinical Biochemistry, Rigshospitalet and Department of Biomedical Sciences, Rigshospitalet, University of Copenhagen, Denmark

⁴Department of Cardiovascular and Renal Research, University of Southern Denmark, Odense, Denmark

Short running title: High-intensity training and type 2 diabetes

Corresponding author: Stefan P. Mortensen, smortensen@health.sdu.dk
University of Southern Denmark, J.B. Winslows Vej 21 3, 5000 Odense C, Tel: (61717040

ClinicalTrials.gov ID no: NCT02001766

AIM

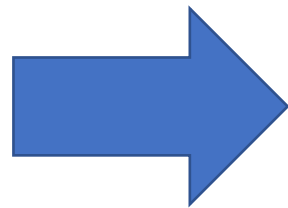
To evaluate if high-intensity interval training (HIIT) with a lower time commitment can be equally effective as endurance training (END) on glycemic control, physical fitness and body composition in individuals with type 2 diabetes.

MATERIALS AND METHODS

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/dom.13198

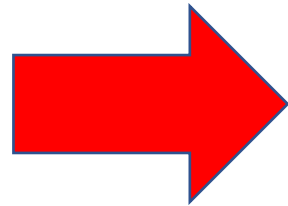
Tenho ao menos 20 - 50 “*seed papers*”?

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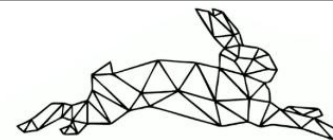


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Effect Of Morning Vs Afternoon Exercise On Continuous Glucose Concentrations In Type 2 Diabetes Patients

Medicine and Science in Sports and Exercise

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2019

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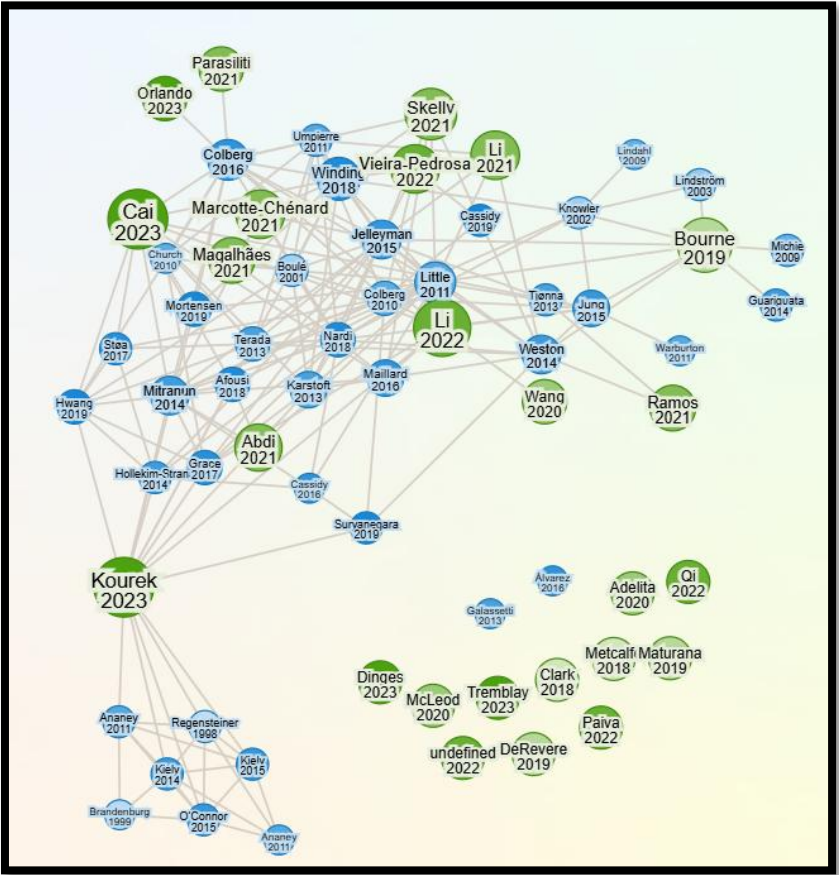
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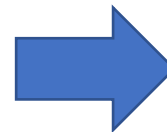
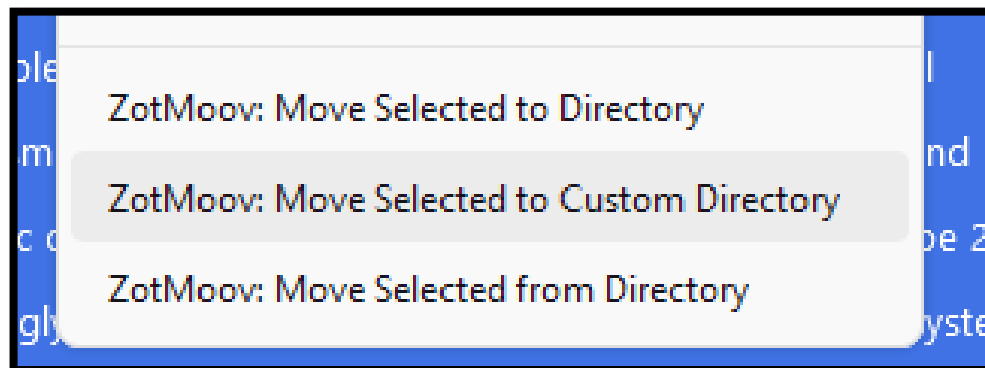
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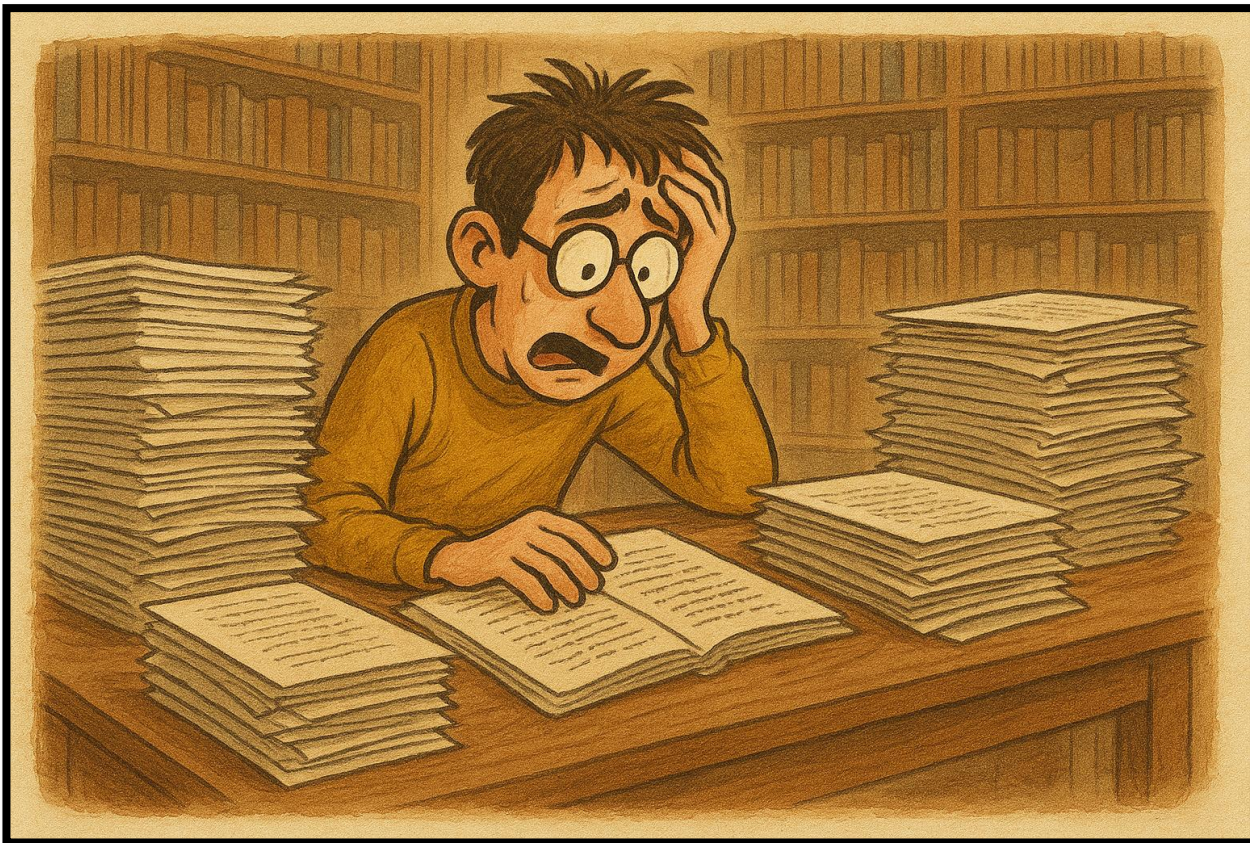
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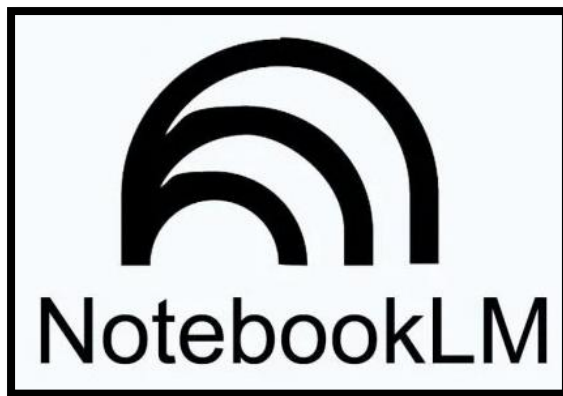
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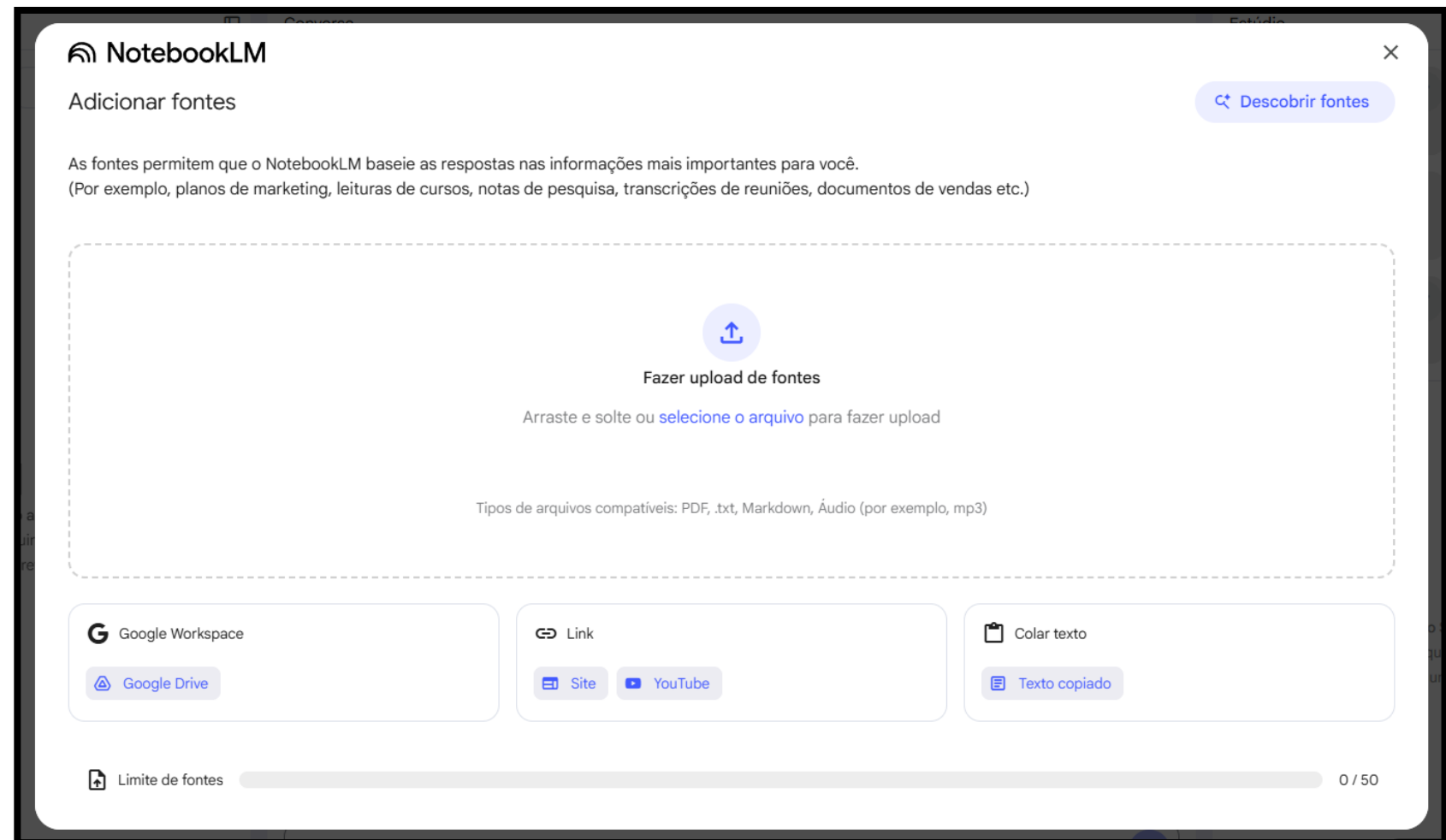


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Interval Training for Cardiometabolic Health: Why Such A HIIT?

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Depois de incluir as fontes, clique para adicionar um Resumo em Áudio, um Guia de Estudo, um Mapa Mental e muito mais!

✎ Adicionar nota

Como o NotebookLM pode ajudar nesta etapa?

Principais lacunas

- Quais são os **principais gaps** identificados pelos autores?
- Quais são as **limitações mais mencionadas** nos estudos?
- Quais são as **principais recomendações** para pesquisas futuras?
- Quais tópicos parecem **subexplorados** ou pouco estudados?

Estado da arte

- Quais são os **principais achados** em comum entre os artigos?
- Quais são as **principais divergências** nos resultados?
- Quais são as **teorias ou hipóteses dominantes** no campo?
- Quais são as **principais aplicações práticas** sugeridas pelos estudos?

Metodologia

- Quais **modelos experimentais** predominam (in vivo, in vitro, humanos)?
- Como os estudos diferem em termos de **desenho de pesquisa** (randomizado, observacional, revisão, meta-análise)?
- Há **tendência de padronização** ou cada estudo usa métodos distintos?
- Quais são as **populações-alvo** (idade, sexo, condição clínica)?

Próximo passo...

Vamos praticar



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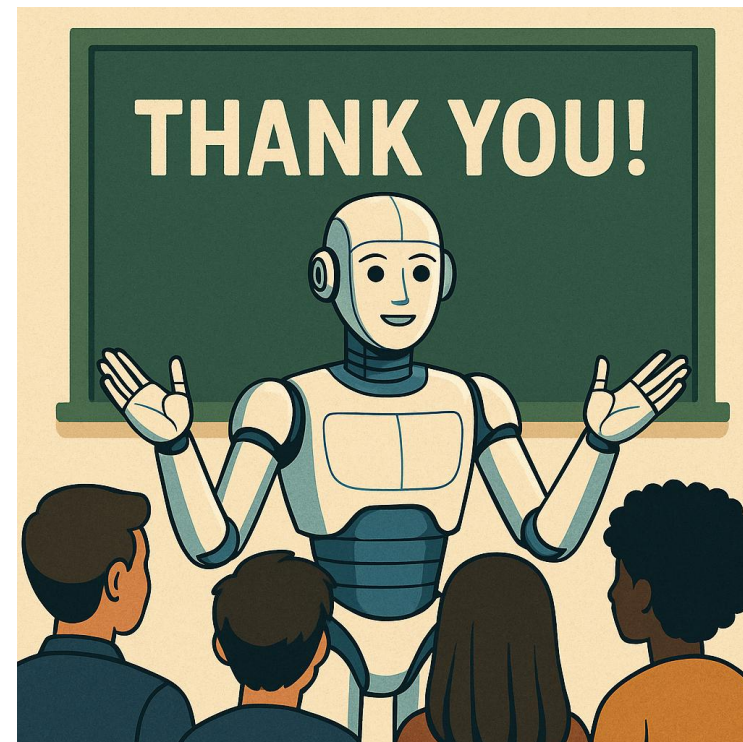
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